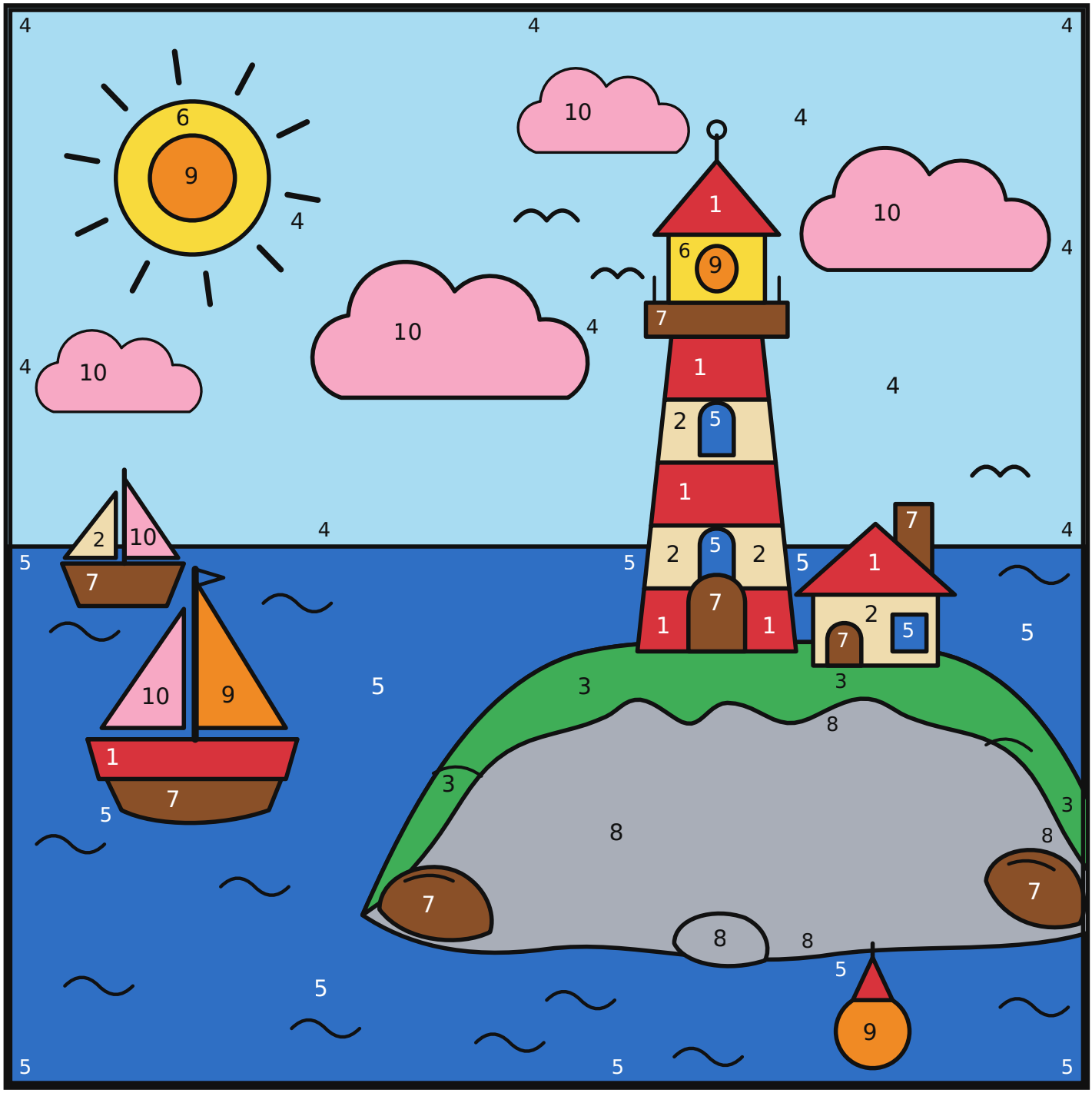
Name: **ANSWER KEY** _____ Date: _____

Functions & Transformations: Color By Number



Color Code:

9 Orange 10	**1** Red 15	**5** Blue 5	**10** Pink 2	**4** Light Blue 9
6 Yellow −6	**2** Tan −3	**8** Gray 11	**7** Brown 7	**3** Green −5

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Directions: Solve each problem and show your work in the box. Your answer should match one of the answers in the Color Code on the coloring sheet. Find the matching answer, write the problem number on its blank, then write the color on the line. Color every part of the picture with that number. Watch closely: the answers look alike!

1.

Let $f(x) = 2x^2 - x$. Find $f(-3) - f(2)$.

$f(-3) = 2(-3)^2 - (-3) = 18 + 3 = 21$
$f(2) = 2(2)^2 - 2 = 6$
$21 - 6$
15

1. Color: **Red**

2.

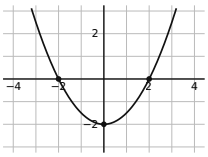
What is the smallest x-value in the domain of $f(x) = \sqrt{(4x + 12)} - 2$?

Need $4x + 12 \geq 0$, so $4x \geq -12$ and $x \geq -3$
The -2 shifts the graph down; it changes the range, not the domain.
−3

2. Color: **Tan**

3.

The graph of $y = f(x)$ is shown. If $h(x) = 3f(x - 2) + 1$, find $h(2)$.



$h(2) = 3f(0) + 1 = 3(-2) + 1$, since $f(0) = -2$
−5

3. Color: **Green**

4.

The point $(-3, 4)$ is on $y = f(x)$. Find the x-coordinate of the matching point on $y = f(-\frac{x}{3})$.

$-\frac{x}{3} = -3$, so $x = (-3)(-3)$
9

4. Color: **Light Blue**

5.

The graph of $y = \sqrt{x}$ is reflected in the x-axis, then shifted 5 units left and 7 units up to make g. Find $g(-1)$.

$g(x) = -\sqrt{(x + 5)} + 7$
$g(-1) = -\sqrt{4} + 7 = -2 + 7$
5

5. Color: **Blue**

6.

Let $f(x) = \begin{cases} 2x + 7 & \text{if } x < -1 \\ x^2 - 5 & \text{if } -1 \leq x \leq 3 \\ 11 - x & \text{if } x > 3 \end{cases}$

Find $f(-1) + f(3) - f(5)$.

$f(-1) = (-1)^2 - 5 = -4$, $f(3) = 4$, $f(5) = 6$
$-4 + 4 - 6$
−6

6. Color: **Yellow**

7.

Let $f(x) = 2x - 3$ and $g(x) = x^2 + 1$. Find $(f \circ g)(-2)$.

$(f \circ g)(-2) = f(g(-2))$: work inside first
$g(-2) = (-2)^2 + 1 = 5$
$f(5) = 2(5) - 3$
7

7. Color: **Brown**

8.

Use the table to find $(f \circ g)(1) + (f \cdot g)(0)$.

x	−2	−1	0	1	2
$f(x)$	3	5	−2	4	1
$g(x)$	0	−2	−3	−1	3

$(f \circ g)(1) = f(-1) = 5$; $(f \cdot g)(0) = (-2)(-3) = 6$
$5 + 6$
11

8. Color: **Gray**

9.

$T(c) = \frac{9}{5}c + 32$ converts °C to °F. Find $T^{-1}(50)$, the °C that equals 50°F.

$T^{-1}(c) = \frac{5}{9}(c - 32)$, so $T^{-1}(50) = \frac{5}{9}(18)$
10

9. Color: **Orange**

10.

A \$60 jacket has a \$10 coupon, $c(p) = p - 10$, and 20% off, $s(p) = 0.8p$. How many dollars cheaper is the better order?

$(s \circ c)(60) = 0.8(50) = 40$; $(c \circ s)(60) = 48 - 10 = 38$
$40 - 38$
2

10. Color: **Pink**